Thapar Institute of Engineering and Technology
Computer Organization and Architecture
B.Tech. Auxiliary Examination

Code: UCS510/UCS319 **Roll No.....**
TM : 100 Marks **Date: 20/8/2024**
Instructor Name- SHI **Time: 3 hours**
Note: All Questions are compulsory and must be answered in the given sequence only.

Q1 A:	Implement the 1×8 multiplexer using a lower order multiplexer. Draw a clean diagram and the truth table showing the combination of select lines and the output.	**(10+10)**
Q1 B:	Simplify the given 5 variable k map F(A, B, C, D, E) = Σ (1, 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31) showing each step and grouping of min-terms clearly.	
Q2 A	Represent the following in single precision IEEE 754 floating-point format. Show each step clearly. a. (-0000.1111) Binary b. (14.625) Decimal	**(10+10)**
Q2 B:	Explain BNZ, BUN, and BSA and INZ Instruction using their micro-operation.	
Q3 A:	Consider a 16-bit processor in which the main memory appears as in Figure (below). The instruction is 32-bit contained by addresses 200 and 201. Determine the effective address and operand to be loaded in the accumulator for the following addressing modes. a. Indirect, b. PC relative, c. Auto increment d. Auto decrement e. Indexed Instruction: Opcode | 250 400 R1 100 RX Main Memory 200: Load to AC | Mode 201: 500 202: Next Instruction 250: 1100 399: 999 400: 1000 500: 1100 600: 1200 702: 1302 1100: 1700	**(10+10)**
Q3 B:	Name the different hazards observed in the pipelining. Explain each of them with example (set of instructions where the particular hazard is occurring)	

Q4A: Q4 B:	Using BOOTH's Algorithm, show the multiplication of the following signed decimal **(-5 X 10)** numbers. The number of bits to represent each digit should be 5. Show each step very clearly. A. Write the difference between microprogrammed and hardwired control Unit B. Write the difference between RISC and CISC	**(10+10)**
Q5 A: Q5 B:	Consider a non-pipelined processor with a clock rate of 3.6 gigahertz and average cycles per instruction of 5. The same processor is upgraded to a pipelined processor with six stages, but due to the internal pipeline delay, the clock speed is reduced to 2.5 gigahertz. Assume there are stalls of 1 ns after each stage in the pipeline. Find the speed up achieved in this pipelined processor. Show each step clearly. What is DMA, explain its working with neat and clean diagram. (consider following points: Diagram, Registers included, step by step working, modes of transfer)	**(10+10)**